Yingxian Li . Jarvis Consulting

I recently graduated from the University of Toronto with a master?s degree in Data Analytics and Machine Learning. Through hands-on projects at Jarvis Consulting Group, I have built data engineering and machine learning solutions using Python, SQL, PySpark, Scikit-learn, Databricks, Azure, and cloud-based data platforms. My experience includes building ETL pipelines, medallion architecture solutions, cloud analytics workflows, financial ML models, and interactive dashboards. I am especially interested in applying data engineering and machine learning to solve practical business problems in cloud environments. My strongest hands-on experience is in Databricks, PySpark, and SQL, and I have also worked with Azure-based Databricks solutions. I bring strong technical fundamentals, fast ramp-up ability, and a fresh mindset to teams modernizing their data and AI solutions.

Skills

Proficient: Python, SQL, Databricks, Linux/Bash, Git

Competent: AWS, Data Modeling, Machine Learning, Scikit-learn, Model Evaluation, ETL/ELT, Agile/Scrum

Familiar: Azure, SHAP, Airflow, Docker, Java

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_YingxianLi

Cluster Monitor [GitHub]:

- Built a Linux-based cluster monitoring agent that collects host CPU/memory/disk metrics and stores them in PostgreSQL.
- Automated metric collection with Bash scripts and scheduled jobs, enabling near real-time visibility into server health.
- Wrote SQL queries to analyze resource usage trends and support capacity planning.

RDBMS and SQL [GitHub]:

- Designed an ERD and implemented a relational schema using PostgreSQL DDL with primary/foreign keys and constraints.
- Wrote optimized SQL queries (joins, aggregates, subqueries, CTEs, window functions) to answer business questions.

Core Java Apps [GitHub]: Grep App: Built a Java CLI app that mimics Linux `grep` by recursively scanning directories, matching lines using regex (`Pattern`), and writing results to an output file; improved scalability by streaming file contents line-by-line instead of loading whole files.

Python Data Analytics [GitHub]:

- Built a Python-based retail analytics proof of concept for the London Gift Shop (LGS) marketing team using Jupyter Notebook.
- Cleaned and transformed historical transaction data to support exploratory analysis and business insight generation.
- Generated marketing-focused insights to help identify high-value, loyal, inactive, and at-risk customers.

Spark [GitHub]:

- Redesigned the earlier Python retail analytics workflow into a scalable Spark-based solution for larger datasets.
- Built the main implementation in Databricks on Azure using PySpark DataFrame APIs for data cleaning, transformation, aggregation, and business analysis.
- Analyzed retail transaction data to support use cases such as revenue calculation, monthly sales trends, and customer behavior analysis.

Databricks [GitHub]:

- Built reusable Databricks data pipeline components using a medallion architecture, creating standardized Bronze, Silver, and Gold processing patterns that can be extended to additional datasets and use cases.
- Implemented automated ingestion, transformation, data-quality validation, and workflow orchestration using PySpark, Lakeflow Declarative Pipelines, Databricks Jobs, and Unity Catalog.
- Developed governed, analytics-ready Delta tables and interactive Databricks dashboards, enabling business users to access trusted datasets and business metrics.

- Independently researched Databricks platform capabilities, troubleshot integration and permission issues, and documented implementation decisions to support future platform adoption.

Home Credit ML Project [GitHub]:

- Built a credit default prediction project using the Home Credit dataset with Python, Pandas, NumPy, and Scikit-learn
- Performed EDA, missing-value analysis, sentinel-value handling, feature engineering, categorical encoding, and model-ready preprocessing.
- Engineered credit-risk features such as credit-to-income ratio, annuity-to-income ratio, external score aggregates, age buckets, and bureau-level aggregations.
- Compared logistic regression and tree-based models using AUROC, Gini, KS, AUPRC, and F1, with SHAP-based explainability and PSI monitoring.

Highlighted Projects

Credit Scoring ML: Built an end-to-end credit default prediction project using the Home Credit dataset, including EDA, missing-value handling, financial feature engineering, Scikit-learn model comparison, financial metric evaluation, SHAP-based explainability, and PSI monitoring.

Stock Analytics Pipeline: Developed a reusable Databricks data platform solution using Lakeflow Declarative Pipelines and a medallion architecture to ingest, validate, standardize, and transform stock API data across Bronze, Silver, and Gold Delta tables. Implemented incremental ingestion, data-quality expectations, deduplication, and automated workflow orchestration to create reliable and governed datasets in Unity Catalog. Engineered reusable analytical features, including 7-, 30-, and 90-day price-change and volume-trend metrics, and delivered interactive Databricks dashboards with dynamic filters for downstream users. Independently researched unfamiliar Databricks capabilities, resolved storage-access and pipeline-design challenges, and documented the solution for future reuse and extension.

Fitness Cloud Data Platform: Joined an in-progress AWS migration project on short notice and independently ramped up on the AWS ecosystem to support delivery after a team member unexpectedly left. Developed modular ETL pipelines with AWS Glue and PySpark to process historical and streaming CSV/JSON data from Amazon S3 into curated Aurora MySQL tables covering customer profiles, class schedules, attendance, engagement, lifecycle segmentation, and churn-prediction features. Orchestrated daily and weekly workflows using EventBridge-triggered AWS Lambda functions, built an interactive Streamlit dashboard to visualize signup trends, popular classes, instructors, and churn patterns, and implemented an Amazon SES?based email campaign system that sent personalized onboarding and re-engagement messages while preventing duplicate communications within 30 days.

Professional Experiences

Data Engineering & Data Science Trainee, Jarvis Consulting Group (2026.01 - Present): Developed software and data engineering projects using Python, SQL, PySpark, Databricks, PostgreSQL, Linux/Bash, Java, and related cloud technologies. Designed and built ETL pipelines, relational database solutions, system-monitoring applications, backend tools, and data analytics workflows to support technical anad business use cases. Automated data processing and operational tasks, implemented data-quality checks, and used Git and Agile practices throughout the development lifecycle. Translated project requirements into working solutions, troubleshot technical issues, and presented analytical results through reports and interactive dashboards.

Data Analyst Intern, VS-Consulting Group (2022.05-2022.08): Collected, cleaned, and transformed client data from multiple sources into structured datasets for reporting and analysis. Automated recurring reporting workflows using Excel VBA and developed Power BI dashboards to improve visibility into business KPIs and reduce manual effort. Built analytical models based on stakeholder requirements, collaborated with consultants and data engineers to deliver data solutions, and presented findings and recommendations to project leads and clients.

Education

University of Toronto (2023-2025), Master of Engineering, Data Analytics and Machine Learning

University of Toronto (2019-2023), Honours Bachelor of Science, Mathematics - Scholarship - University of Toronto Dean List (2020-2022) - GPA: 3.7/4.0

Miscellaneous

- Databricks Certified Data Engineer Professional (2025)
- Databricks Certified Data Engineer Associate (2025)
- Microsoft Power BI Data Analyst (PL-300) (2022)
- SAS Advanced Programmer Certification (2021)
- SAS Base Programmer Certification (2021)